

AI-Driven Transformation of Business Decision Making and Operations

Rahul Soni

rsoni30@uis.edu

MIS 577 - MIS Design/Research Method

Supervised by

Dr. Roya Gholami

Associate Professor, Management Information Systems

August 2023

Abstract

The growing use of artificial intelligence in business is expanding the ability of organizations to automate processes and make decisions that have traditionally depended on human involvement. However, increasing technical capability does not by itself determine whether a business decision should be performed autonomously. This study examines the relationship between AI-based decision making, business autonomy, and human oversight through a review of relevant literature together with an experimental analysis of machine-learning-based credit-risk classification.

Five supervised machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Gradient Boosting, were evaluated using a credit-risk dataset containing 1,000 observations. Random Forest achieved the highest overall accuracy of 76.0 percent, while Support Vector Machine achieved the highest ROC-AUC of 0.7819. Error and threshold analyses showed that aggregate predictive performance can conceal important business risks. At the default classification threshold, Random Forest incorrectly classified 34 of 60 actual bad-risk cases as good risk, while changing the threshold produced substantial trade-offs between precision, recall, false positives, and false negatives.

A selective automation and human-review configuration was subsequently evaluated in which more decisive predictions were processed automatically and less decisive predictions were designated for human review. The configuration allowed AI to handle 67.5 percent of test cases automatically, while 32.5 percent were designated for human review. Accuracy among the AI-handled cases was 82.22 percent, although prediction errors remained.

The findings indicate that the appropriate level of AI autonomy cannot be determined from model accuracy alone. Predictive uncertainty, error consequences, organizational risk, business value, accountability, and governance should also be considered when determining how decision authority is allocated between humans and AI. The study therefore approaches AI-driven business

operations as a continuum in which the level of human involvement is adjusted according to predictive reliability, decision consequences, and organizational requirements.

1 Introduction

Artificial intelligence has moved beyond specialized computational applications and has become increasingly relevant to everyday business operations. Organizations use AI to analyze large volumes of data, identify patterns, generate predictions, automate repetitive activities, and support decisions that have traditionally required substantial human involvement. Machine learning, in particular, enables systems to learn patterns and relationships from data rather than depending entirely on rules specified in advance, creating new possibilities for prediction, automation, and organizational decision making (Jarrahi, 2018; Wamba-Taguimdje et al., 2020). At the organizational level, however, the value of AI does not come from algorithms and technical infrastructure alone. It also depends on how firms combine technology with data, employees, business processes, managerial capabilities, and other organizational resources (Schmidt et al., 2020; Mikalef and Gupta, 2021).

As AI becomes more deeply embedded in business processes, an important question is how decision authority should be divided between people and technology. Information systems have long supported managers by collecting, organizing, and processing information, while responsibility for interpreting that information and making decisions has largely remained with people. Advances in AI and machine learning have changed this relationship by allowing parts of the decision process itself to be automated, particularly when decisions involve prediction, classification, pattern recognition, or the analysis of large quantities of data (Berente et al., 2021; Agrawal et al., 2019). This does not necessarily imply the removal of people from decision making. Instead, AI creates different ways of organizing work, ranging from the substitution of particular human tasks to arrangements in which human and AI capabilities complement one another (Rai et al., 2019).

The distinction between prediction and decision mak-

ing is especially important. Machine learning can produce a prediction about an uncertain outcome, but that prediction does not by itself determine what action should be taken. A business decision may also require judgment about objectives, costs, benefits, risks, and the consequences of different actions. Agrawal et al. (2019) distinguish prediction from the broader decision process and show that improvements in machine prediction can either automate particular tasks or increase the value of complementary human judgment. This distinction helps explain why greater AI capability does not automatically result in complete decision automation.

The different strengths of humans and AI become increasingly important as businesses consider giving automated systems greater responsibility for operational activities. AI can process large quantities of information rapidly and consistently, whereas human decision makers can contribute contextual understanding, judgment, creativity, and the ability to interpret circumstances that may not be adequately represented in available data. Human-AI collaborative decision making should therefore not be understood simply as a choice between a person and an algorithm. The appropriate allocation of decision tasks depends on the nature of the problem and the relative capabilities of the human and algorithmic decision makers involved. Some tasks may be better suited to algorithms, others may continue to require human judgment, and some may benefit from combining both forms of decision making (Puranam, 2021).

These differences create opportunities for organizations to automate activities that extend beyond simple and repetitive processes. AI can contribute to customer service, forecasting, risk assessment, classification, planning, resource allocation, information analysis, and other operational decisions (Patel and Trivedi, 2020; Gkikas and Theodoridis, 2022). AI is also extending automation beyond routine and repetitive tasks, allowing machines to support or replace some forms of non-routine decision making (Zhang et al., 2018). Demand forecasting provides one example, where machine learning can use historical business data to predict future demand and support inventory, purchasing, and operational planning decisions (Khan et al., 2020). Research examining digital labor platforms describes different arrangements through which AI can substitute for particular human tasks, augment human capabilities, or become integrated with human work (Rai et al., 2019). Similar developments can be observed in accounting and auditing, where AI and related technologies can automate information processing and analytical activities while professional judgment and responsibility remain important for tasks involving interpretation and consequential decisions (Hasan, 2022). These examples suggest that business automation is better understood in terms of different levels of human involvement rather than as a simple choice between fully human and fully automated operations.

The ability to automate a business activity, however, does not mean that doing so will necessarily create business value. Organizations can make substantial investments in digital technologies without obtaining the ex-

pected operational or financial returns. Research on digital business models describes this problem through the digitalization paradox, in which investment in technologies such as AI does not automatically lead to successful commercialization or improved financial performance (Linde et al., 2021). Research examining AI investments similarly emphasizes the importance of connecting technological initiatives with organizational problems, strategic objectives, desired outcomes, and the capabilities required to implement them effectively (Gudigantala et al., 2023). The business value of greater AI-based automation must therefore be considered alongside the technical capability to automate a particular task.

This relationship becomes more complex when AI systems are used in decisions that have meaningful consequences for customers, employees, organizations, or other stakeholders. Machine learning systems depend heavily on the data used to develop and operate them, and weaknesses in data quality or model design can produce inaccurate, biased, or inappropriate outcomes. Automated decisions can also make it more difficult to understand why a particular outcome occurred or who should be held responsible for it. Research on intelligent automation identifies human judgment, governance, accountability, and the allocation of decision control between people and automated systems as important organizational concerns (Coombs et al., 2020). As direct human involvement decreases, organizations must determine not only whether a decision can be automated, but also whether it should be.

Accountability is particularly important when decision authority is delegated to AI. Allowing an automated system to perform part of a decision process does not necessarily remove the organization's responsibility for the resulting outcome. Research on AI-based organizational decision making has examined different structures for allocating decision authority between humans and AI and has highlighted the difficulty of separating delegated decision authority from organizational accountability (Shrestha et al., 2019). Ethical research on algorithms raises related concerns about lack of transparency, bias, discrimination, privacy, autonomy, and the ability to trace responsibility for algorithmic outcomes (Tsamados et al., 2022). These concerns suggest that increasing operational autonomy must be accompanied by mechanisms that allow organizations to maintain appropriate oversight and responsibility.

Existing research provides substantial insight into individual parts of this development. Studies have examined AI capabilities and organizational value, machine prediction, human-AI collaboration, automated decision making, digital business models, AI implementation, and the ethical and governance implications of algorithmic systems. However, these areas are often considered separately. Research on intelligent automation points to continuing questions about how organizations should allocate decision control between humans and automated systems and how governance arrangements should be designed as automation becomes more extensive (Coombs et al., 2020). At the same time, research on AI investment shows that organizations continue to

face difficulties in translating technological capability into measurable and sustainable business value (Gudigantala et al., 2023). Together, these issues indicate a need to consider AI-based automation not only as a technological development, but also as a business and organizational decision about how much responsibility should be assigned to AI and how much should remain with people.

To address this gap, this paper brings together research on AI-based decision making, human-AI collaboration, machine-learning-enabled automation, business value creation, and responsible governance. It examines business decision making as a continuum in which AI can support human judgment, perform particular decision tasks, or allow some operational activities to proceed with limited direct human involvement. The review considers how the appropriate level of automation may depend on the nature of the task, the quality and availability of data, organizational capabilities, operational risk, expected business value, and the consequences associated with a decision.

2 Related Work

Research on artificial intelligence in business has developed across several connected areas, including machine-learning-based decision support, human-AI decision making, process automation, organizational capabilities, business value, and responsible governance. These areas become closely connected when AI moves beyond providing information and begins to assume greater responsibility within business decisions and operational processes. The literature reviewed in this section provides the foundation for examining how decision authority can be distributed between humans and AI and what conditions should influence the level of automation adopted by an organization.

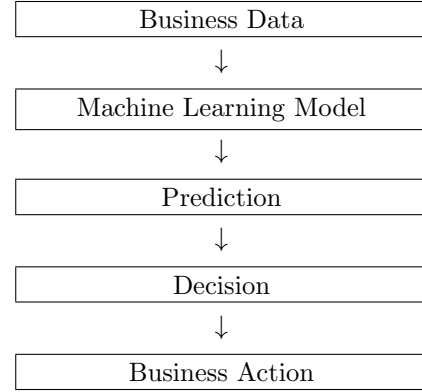
2.1 AI and Machine Learning in Business Decision Making

Machine learning provides the technical foundation for many AI-based business applications by enabling systems to identify patterns in data and generate predictions without requiring every decision rule to be explicitly programmed. These capabilities have been applied to classification, forecasting, fraud detection, risk assessment, customer analysis, and other decision-related activities. Hasan (2022), for example, describes applications of machine learning and related AI techniques in accounting and auditing, including transaction classification, fraud detection, bankruptcy prediction, document analysis, and risk assessment. Machine-learning-based forecasting can also support operational decision making by converting historical sales and inventory data into estimates of future demand that can inform production, purchasing, and inventory planning (Khan et al., 2020).

An important distinction exists between prediction and decision making. Agrawal et al. (2019) describe advances in AI largely as improvements in prediction

technology. A prediction estimates an uncertain outcome using available information, whereas a business decision also requires consideration of possible actions, objectives, costs, benefits, and consequences. Automating a prediction task therefore does not necessarily mean that the resulting business decision should also be automated.

This relationship can be represented as:



The prediction produced by a model may inform a human decision, contribute to a shared human-AI process, or trigger an automated action. Which arrangement is appropriate depends on the business context and the consequences associated with the decision (Volkmar et al., 2022).

Gudigantala et al. (2023) further emphasize that AI initiatives should begin with the business problem, desired outcomes, performance measures, and available data rather than with the selection of an algorithm. Model performance is therefore only one component of AI-based business decision making. Organizations must also determine how predictions are interpreted and converted into actions that contribute to meaningful business outcomes.

2.2 Human and AI Decision Making

The increasing capability of AI raises the question of how decision authority should be distributed between human and algorithmic decision makers. Possible arrangements range from AI systems that provide information or recommendations to humans, to collaborative structures in which humans and AI contribute jointly, and to situations in which greater decision authority is delegated to an automated system.

Puranam (2021) approaches human-AI collaborative decision making as an organization design problem. Human decision makers can contribute contextual understanding and broader judgment, while algorithms can analyze large volumes of information consistently and identify patterns in data. The appropriate arrangement therefore depends on the characteristics of the task and on how human and algorithmic capabilities are combined.

Shrestha et al. (2019) similarly examine alternative structures for distributing decision authority between humans and AI. Their work includes greater delegation to AI, hybrid human-AI decision making, and arrangements that combine human and algorithmic judgments.

These structures demonstrate that AI adoption does not lead to a single form of organizational decision making.

Research on intelligent automation also shows that AI can both substitute for and complement human work (Coombs et al., 2020). Structured and predictable tasks may permit greater automation, whereas decisions involving interpretation, contextual knowledge, or judgment may justify continued human involvement (Volkmar et al., 2022). AI can therefore change how decision authority is allocated without necessarily removing humans from the decision process.

2.3 AI-Enabled Automation and Business Operations

AI moves beyond decision support when predictions or recommendations are connected directly to operational actions. In these situations, AI may perform parts of a process that were previously carried out manually. Coombs et al. (2020) distinguish between automation, in which technology performs a task, and augmentation, in which technology complements human capabilities.

Rai et al. (2019) describe human-AI work through task substitution, task augmentation, and task assemblage. In substitution, AI performs a task previously undertaken by a person. In augmentation, AI complements human capabilities. Task assemblage involves a more integrated arrangement in which human and AI capabilities contribute jointly to the performance of work.

These distinctions are important because a single business process can contain activities with different requirements. Classification, information processing, forecasting, or predefined operational actions may permit substantial automation, while other activities may continue to require interpretation, contextual understanding, or professional judgment. Evidence from accounting and auditing illustrates this distinction, where AI can improve information processing and analytical activities while human judgment remains important for consequential decisions (Hasan, 2022).

Automation can also affect workflows, employee responsibilities, skills, management practices, and organizational coordination (Ris et al., 2020). Berente et al. (2021) note that AI creates opportunities for new products, services, business models, and organizational forms while also introducing challenges involving human-AI interaction, data, privacy, security, ethics, labor, and accountability. Operational autonomy therefore depends not only on whether AI can perform a task but also on whether the surrounding organization can support and govern that automation.

2.4 AI Capability, Business Value, and Business Models

The ability to automate a task does not establish that doing so will create business value. Mikalef and Gupta (2021) describe AI capability as an organization's ability to select, coordinate, and deploy AI-specific resources together with other organizational resources. Obtaining

value from AI therefore requires more than algorithms and technical infrastructure. It also depends on data, human expertise, managerial capabilities, and organizational processes (Mikalef and Gupta, 2021).

AI may create value through improved predictions, faster decisions, increased efficiency, reduced operating costs, or new products and services (Mishra and Tripathi, 2021). Organizations must nevertheless convert these improvements into economic or strategic benefits. Greater automation is therefore not valuable simply because it reduces direct human involvement.

Reim et al. (2020) examine how AI can influence value creation, value delivery, and value capture within business models. Linde et al. (2021) similarly show that technological opportunities do not automatically produce viable business models. Their analysis identifies difficulties that can arise when organizations develop technologically sophisticated solutions without establishing sufficient customer value, delivery capability, or a sustainable mechanism for capturing value. AI can also contribute to business-model transformation by integrating data-driven intelligence, automation, digital platforms, and business strategy (Mishra and Tripathi, 2021).

The economic benefits of AI may also depend on complementary organizational changes. Brynjolfsson et al. (2017) suggest that the benefits of general-purpose technologies may take time to emerge while organizations develop complementary innovations, acquire skills, redesign processes, and reorganize work.

Gudigantala et al. (2023) similarly emphasize the importance of connecting AI investment with business strategy and clearly defined outcomes. In the context of greater operational autonomy, these findings indicate that decisions to increase automation should be justified by improvements in relevant business outcomes rather than by technical capability alone.

2.5 Governance, Accountability, and Responsible AI

As AI assumes greater responsibility within business decisions, governance becomes increasingly important. Organizations need to consider whether automated outputs can be understood, whether the underlying data and predictions are sufficiently reliable and fair, and who remains accountable for actions taken on the basis of those predictions.

Transparency is one concern. Tsamados et al. (2022) identify the difficulty of understanding algorithmic evidence as an important ethical issue. When the basis of an automated decision is unclear, identifying errors, challenging inappropriate outcomes, and establishing responsibility can become more difficult.

Data quality and bias create related concerns because machine learning models learn from available data. Weaknesses or biases in those data can influence subsequent predictions and may contribute to inappropriate or discriminatory outcomes (Tsamados et al., 2022; Stahl et al., 2021). These concerns become particularly important when AI contributes to consequential decisions such

as risk assessments, applications, access to services, or the allocation of opportunities. Although AI can reduce some forms of human decision bias, models may also learn and amplify biases contained in historical data, creating additional governance concerns (Zhang et al., 2018).

Accountability presents an additional challenge. Shrestha et al. (2019) show that decision authority can be distributed between humans and AI in different ways, but greater delegation to AI does not necessarily transfer organizational responsibility for the resulting outcomes. A system may be technically capable of generating a prediction and initiating an action without human approval, but technical capability alone does not determine whether such independence is appropriate.

Coombs et al. (2020) similarly identify human judgment, decision control, accountability, and governance as continuing concerns in intelligent automation. Decisions with substantial financial, legal, ethical, or social consequences may require stronger human involvement, whereas routine and lower-risk activities may permit greater automation when appropriate monitoring and intervention mechanisms are available.

Responsible AI therefore involves both technical and organizational controls. Model evaluation, data governance, documentation, auditing, clearly assigned responsibilities, and mechanisms for human intervention can help organizations retain appropriate oversight as the level of AI autonomy increases (Kumar et al., 2020; Stahl et al., 2021).

2.6 Research Gap

Existing research explains several important components of AI-based business decision making. Research on machine prediction examines how predictions can be generated from data, while human-AI research considers different ways of allocating decision tasks between people and algorithms (Agrawal et al., 2019; Puranam, 2021). Research on intelligent automation examines how AI can assume operational activities, and organizational studies identify the capabilities needed to create value from AI (Coombs et al., 2020; Mikalef and Gupta, 2021). Governance research additionally shows that greater decision authority introduces concerns involving reliability, accountability, transparency, and human oversight (Shrestha et al., 2019; Tsamados et al., 2022).

These perspectives, however, do not by themselves establish how much decision authority should be assigned to AI in a particular business process. Strong predictive performance does not necessarily imply that every prediction should be converted directly into an automated action. The consequences of different prediction errors, the uncertainty associated with individual predictions, the value created by automation, and the need for accountability can all affect whether a particular case should be handled automatically or remain subject to human review.

This issue is particularly relevant when model performance is evaluated using aggregate measures such as accuracy. Two models with similar overall performance

may produce different types of errors, and changing the classification threshold can alter the balance between false positives and false negatives. In a business setting, these errors may have different consequences. Evaluating AI autonomy therefore requires attention not only to how frequently a model is correct but also to where it makes errors and how confidently individual decisions can be made.

The literature therefore leaves the connection between predictive performance and operational decision authority insufficiently examined. Rather than treating business processes as either fully human or fully automated, AI-based decision making can be considered as a continuum in which some predictions are processed automatically while others receive human review. The appropriate position on this continuum depends on predictive reliability, uncertainty, consequences of error, organizational risk, expected business value, and accountability requirements.

This study addresses that connection by combining the organizational literature with an experimental analysis of machine-learning-based credit-risk classification. Model comparison, error analysis, classification-threshold analysis, and a selective automation and human-review configuration are used to examine how predictive behavior affects the level of decision responsibility that can reasonably be assigned to AI.

Based on this gap, the study addresses the following research questions:

- RQ1.** How does machine-learning performance influence the extent to which business decisions can be assigned to automated processing?
- RQ2.** What technical, organizational, and business factors influence the appropriate level of human involvement in AI-based business decision making?
- RQ3.** How can organizations increase AI-based decision autonomy while accounting for predictive uncertainty, consequences of error, business value, accountability, and human oversight?

3 Research Methodology

This study combines a focused literature review with an empirical machine-learning experiment to examine how predictive performance, classification errors, and prediction confidence can influence the allocation of decision authority between AI and human review. The literature component establishes the organizational factors relevant to AI-based decision autonomy, while the experimental component examines these issues in a credit-risk classification setting.

3.1 Research Design

The study uses two complementary components. First, relevant literature is synthesized to examine AI-based decision making, human-AI collaboration, automation, business value, and responsible governance. Second, a

supervised machine-learning experiment is conducted to evaluate how model performance and prediction behavior affect the suitability of automated decision processing.

The experiment does not treat predictive accuracy as equivalent to decision autonomy. Instead, model performance, classification errors, decision thresholds, and prediction confidence are examined separately. This allows the technical performance of the models to be considered alongside the broader question of whether particular predictions should be processed automatically or designated for human review.

3.2 Literature Selection and Analysis

Literature was collected through targeted searches using Google Scholar and Google. Search terms covered artificial intelligence in business, machine learning for decision making, human-AI collaboration, intelligent automation, algorithmic decision making, AI business value, human oversight, accountability, governance, and responsible AI.

Publications were selected according to their relevance to the central themes of the study. The literature was analyzed thematically rather than by summarizing each publication independently. Five themes were used: AI and machine learning in business decision making, human-AI decision making, AI-enabled automation and business operations, AI capability and business value, and governance, accountability, and responsible AI.

The resulting synthesis was used to identify factors that may influence the level of decision authority assigned to AI, including predictive reliability, data quality, task characteristics, business value, consequences of error, organizational risk, and accountability requirements.

3.3 Experimental Dataset

The empirical analysis uses the German Credit Risk dataset, which contains 1,000 observations. Each observation represents a credit case described by demographic and financial attributes.

The dataset contains the following variables: age, sex, job category, housing status, saving-account status, checking-account status, credit amount, credit duration, purpose of the credit, and credit risk.

The target variable, *Risk*, originally contains two categories: *good* and *bad*. For the experiment, the target was encoded as:

$$\text{good risk} = 0, \quad \text{bad risk} = 1.$$

The dataset contains 700 good-risk observations and 300 bad-risk observations, producing an imbalanced binary classification problem in which the bad-risk class represents 30 percent of the observations.

3.4 Data Preprocessing

The dataset contained an unnamed index column that did not represent a predictive feature and was removed

before model development.

Missing values were present in the *Saving accounts* and *Checking account* variables. These missing categorical values were represented using an *unknown* category rather than removing the affected observations. After preprocessing, no missing values remained.

The predictor variables consisted of both numerical and categorical attributes. Categorical variables were transformed using one-hot encoding. Numerical features were standardized where required for algorithms sensitive to differences in feature scale, particularly Logistic Regression and Support Vector Machine.

The data were divided into training and testing sets using an 80-20 split with stratification on the target variable. The training set contained 800 observations, including 560 good-risk and 240 bad-risk cases. The test set contained 200 observations, including 140 good-risk and 60 bad-risk cases. Stratification preserved the original class distribution across the two sets. All preprocessing transformations were fitted using the training data and subsequently applied to the test data to prevent information from the test set from being used during model training.

3.5 Machine Learning Models

Five supervised classification algorithms were evaluated:

1. Logistic Regression,
2. Decision Tree,
3. Random Forest,
4. Support Vector Machine (SVM), and
5. Gradient Boosting.

The purpose of comparing multiple algorithms was not only to identify the model with the highest predictive performance but also to examine differences in the types of classification errors produced by the models. These models were selected to provide comparisons across linear, tree-based, ensemble, margin-based, and boosting approaches.

3.6 Model Evaluation

Model performance was evaluated on the held-out test set using accuracy, precision, recall, F1 score, and receiver operating characteristic area under the curve (ROC-AUC).

For all classification metrics, the bad-risk class was treated as the positive class.

Accuracy measures the proportion of all test observations classified correctly:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Precision measures the proportion of cases predicted as bad risk that were actually bad risk:

$$\text{Precision} = \frac{TP}{TP + FP}.$$

Recall measures the proportion of actual bad-risk cases correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}.$$

The F1 score provides the harmonic mean of precision and recall:

$$F1 = 2 \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right).$$

ROC-AUC was additionally used to evaluate the ability of each model to distinguish between the two risk classes across classification thresholds.

Because the business consequences of classification errors may differ, confusion matrices were also examined. Particular attention was given to false negatives, representing actual bad-risk cases classified as good risk, and false positives, representing actual good-risk cases classified as bad risk.

3.7 Classification Threshold Analysis

Random Forest produced the highest overall accuracy among the evaluated models and was therefore used for further threshold analysis.

The default binary classification threshold of 0.50 was compared with thresholds of 0.30, 0.40, 0.60, and 0.70. At each threshold, accuracy, precision, recall, F1 score, false positives, false negatives, true positives, and true negatives were calculated.

The purpose of this analysis was to determine how changing the threshold alters the balance between different types of classification errors. Lower thresholds classify a greater number of observations as bad risk and can increase recall while also increasing false positives. Higher thresholds require stronger model confidence before assigning the bad-risk class and can reduce false positives while increasing false negatives.

This analysis is important for business decision making because the default threshold does not necessarily represent the most appropriate operating point when the consequences of the two types of error differ. The threshold values were examined as alternative operating points rather than as optimized decision thresholds.

3.8 Selective Automation and Human Review

A selective automation configuration was constructed to examine whether prediction confidence could be used to determine which cases were processed automatically and which were designated for human review.

Rather than forcing the model to make an automated decision for every observation, predictions with stronger model confidence were assigned directly to an automated decision category, while less decisive predictions were placed in a human-review category.

The selective decision-routing process contained three possible outcomes:

AI: Low Risk → *Human Review* → *AI: High Risk*

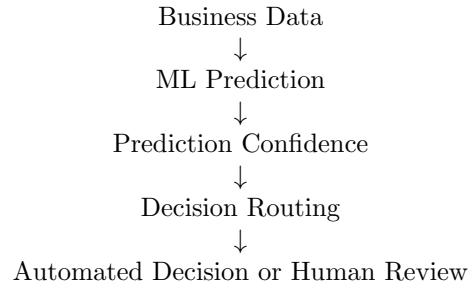
This configuration was evaluated using the number and proportion of cases handled automatically, the number and proportion designated for human review, accuracy among AI-handled cases, and the number of actual bad-risk cases assigned to human review.

The purpose of the selective configuration was not to claim that human reviewers were experimentally observed. No human participants were involved in the experiment. Instead, *Human Review* represents a decision-routing category indicating cases for which the model would defer automated processing.

3.9 Analytical Connection to AI Decision Autonomy

The experimental design treats AI decision autonomy as a continuum rather than a binary distinction between manual and fully automated decision making.

The relationship examined can be summarized as:



Model accuracy provides evidence about overall predictive reliability, while confusion-matrix analysis identifies the types of errors that remain. Threshold analysis demonstrates how business decision rules can alter those errors, and selective automation examines whether prediction confidence can be used to limit automatic processing to a subset of cases.

The experimental results are subsequently interpreted together with the organizational factors identified in the literature, including consequences of error, business value, operational risk, accountability, and the need for human oversight.

4 Experimental Analysis

4.1 Business Decision Problem

The experimental analysis uses credit-risk classification as a representative business decision problem. Credit decisions require organizations to evaluate available customer information and determine whether an applicant represents an acceptable or unacceptable level of financial risk. This setting is useful for examining AI-based decision making because machine learning can generate risk predictions from customer attributes, while the resulting business decision can involve different levels of automated processing and human review.

The objective of the experiment is to compare the predictive behavior of several machine learning models and examine how classification errors, decision thresholds, and prediction confidence affect the suitability of

automated decision processing. The experiment therefore considers not only overall predictive performance but also the types of errors produced by the models.

Credit-risk classification also provides a setting in which different prediction errors have different interpretations. A false positive occurs when an actual good-risk case is classified as bad risk, whereas a false negative occurs when an actual bad-risk case is classified as good risk. Examining these errors separately is important because overall accuracy alone does not show how the model distributes its mistakes between the two classes.

4.2 Model Comparison

Five supervised machine learning algorithms were compared under the same training and testing conditions: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Gradient Boosting. These models represent different approaches to classification and allow differences in predictive behavior to be examined within the same credit-risk problem.

Logistic Regression provides a linear baseline for comparison. Decision Tree classifies observations through a sequence of feature-based decision rules and offers a relatively interpretable representation of the classification process. Random Forest combines predictions from multiple decision trees and can improve predictive stability compared with a single tree (Abdulkareem and Abdulazeez, 2021). Support Vector Machine identifies a decision boundary that separates observations from the two classes, while Gradient Boosting builds an ensemble sequentially by giving greater attention to errors produced by earlier learners.

The comparison focuses on accuracy, precision, recall, F1 score, ROC-AUC, and the distribution of classification errors. Particular attention is given to the bad-risk class because incorrectly classifying an actual bad-risk case as good risk may create different business consequences from incorrectly classifying a good-risk case as bad risk.

4.3 Threshold-Based Decision Analysis

Following the initial model comparison, Random Forest was selected for additional analysis because it achieved the highest overall test accuracy. The default classification threshold of 0.50 was compared with alternative thresholds of 0.30, 0.40, 0.60, and 0.70.

The purpose of varying the threshold was not to identify a universally optimal operating point. Instead, the analysis examines how changes in the classification boundary alter the balance between false positives and false negatives. Lower thresholds make the model more likely to classify cases as bad risk, which can improve recall but may also increase the number of good-risk cases incorrectly classified as bad risk. Higher thresholds create the opposite trade-off.

This analysis connects model prediction with business decision policy because the threshold determines how predicted probabilities are translated into classifications. The appropriate threshold therefore depends not

only on statistical performance but also on the relative consequences of the two types of classification error.

4.4 Selective Automation Analysis

The final stage of the analysis examines whether prediction confidence can be used to allocate different levels of decision authority between automated processing and human review.

Rather than requiring the model to generate an automated decision for every case, the selective configuration separates predictions into three decision categories:

$$AI: Low Risk \rightarrow Human Review \rightarrow AI: High Risk$$

Cases with stronger model confidence are assigned to an automated category, while less decisive predictions are designated for human review. This creates an intermediate level between complete manual processing and complete automation.

The purpose of the selective configuration is to examine whether prediction confidence can support decision routing within an AI-based business process. The human-review category represents a routing mechanism rather than experimentally observed human decision performance. No human participants were involved in the experiment.

4.5 Analytical Focus

The analysis therefore considers three related questions. First, how do different machine learning models vary in predictive performance and classification errors? Second, how does changing the classification threshold alter the types of errors produced by the selected model? Third, how can prediction confidence be used to determine which cases are processed automatically and which are retained for human review?

Together, these analyses connect technical model behavior with the broader question of how much operational decision authority can reasonably be assigned to AI.

5 Experimental Results

5.1 Model Performance Comparison

The five machine learning models produced different trade-offs between overall predictive accuracy and their ability to identify bad credit-risk cases. Table 1 presents the performance of each model using accuracy, precision, recall, F1 Score, and ROC-AUC.

Table 1: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
SVM	0.745	0.6452	0.3333	0.4396	0.7819
Random Forest	0.760	0.6500	0.4333	0.5200	0.7751
Gradient Boosting	0.730	0.5750	0.3833	0.4600	0.7633
Logistic Regression	0.735	0.5946	0.3667	0.4536	0.7611
Decision Tree	0.710	0.5172	0.5000	0.5085	0.6500

Random Forest achieved the highest overall accuracy at 76.0 percent and the highest F1 Score at 0.5200, while SVM achieved the highest ROC-AUC at 0.7819. Decision Tree produced the highest recall at 50.0 percent, indicating that it identified a larger proportion of the actual bad credit-risk cases than the other models. However, its higher recall was accompanied by lower overall accuracy and precision.

Figure 1 provides a visual comparison of the five evaluation measures across the models.

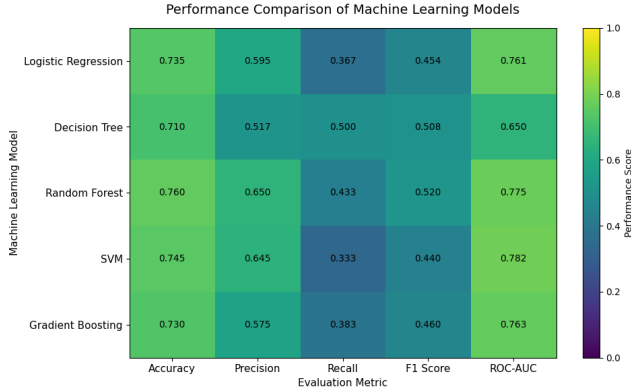


Figure 1: Performance comparison of the five machine learning models.

The results show that no single performance measure is sufficient for determining the suitability of a machine learning model for business decision making. Although Random Forest achieved the highest overall accuracy, its recall of 43.33 percent indicates that more than half of the actual bad credit-risk cases were not identified at the default classification threshold. This distinction is particularly important in credit-risk decisions because different types of classification errors may have different business consequences.

5.2 Error Analysis

The confusion matrix analysis provides a more detailed view of the classification errors produced by each model. Table 2 reports the number of true negatives, false positives, false negatives, and true positives for each model on the 200-observation test set.

Table 2: Confusion Matrix Results for the Machine Learning Models

Model	TN	FP	FN	TP
Logistic Regression	125	15	38	22
Decision Tree	112	28	30	30
Random Forest	126	14	34	26
SVM	129	11	40	20
Gradient Boosting	123	17	37	23

Random Forest correctly classified 126 of the 140 good-risk cases and 26 of the 60 bad-risk cases. However, it incorrectly classified 34 actual bad-risk cases as good

risk. SVM produced the fewest false positives, with 11 good-risk cases incorrectly classified as bad risk, but it also produced the highest number of false negatives, incorrectly classifying 40 bad-risk cases as good risk. Decision Tree produced the fewest false negatives, with 30, but generated the highest number of false positives, with 28.

These results demonstrate that the models differ not only in overall predictive performance but also in the types of errors they produce. In a credit-risk setting, a false positive may result in a good-risk applicant being classified as bad risk, whereas a false negative may result in a bad-risk applicant being classified as good risk. The business consequences of these two errors may differ substantially.

The error analysis therefore reinforces the importance of considering the consequences of classification errors when determining whether an AI system should be given greater decision authority. A model with higher overall accuracy may not necessarily be preferable if it produces a type of error that creates greater financial or operational risk for the organization.

5.3 Classification Threshold Analysis

Because the default classification threshold of 0.50 does not necessarily reflect the relative business consequences of different credit-risk errors, an additional threshold analysis was conducted using the Random Forest model. Classification thresholds ranging from 0.30 to 0.70 were evaluated to examine how changing the threshold affects accuracy, precision, recall, F1 Score, false positives, and false negatives.

Table 3: Random Forest Performance Across Classification Thresholds

Threshold	Accuracy	Precision	Recall	F1 Score	FP	FN
0.30	0.685	0.4831	0.7167	0.5772	46	17
0.40	0.735	0.5507	0.6333	0.5891	31	22
0.50	0.760	0.6500	0.4333	0.5200	14	34
0.60	0.750	0.7083	0.2833	0.4048	7	43
0.70	0.750	0.9167	0.1833	0.3056	1	49

Lowering the classification threshold increased the model’s ability to identify bad-risk cases. At a threshold of 0.30, recall increased to 71.67 percent, compared with 43.33 percent at the default threshold of 0.50. The number of false negatives decreased from 34 to 17. However, this improvement was accompanied by an increase in false positives from 14 to 46, while overall accuracy decreased from 76.0 percent to 68.5 percent.

A threshold of 0.40 produced a comparatively stronger balance between precision and recall among the evaluated thresholds. Recall increased to 63.33 percent, false negatives decreased to 22, and the F1 Score increased to 0.5891, the highest F1 Score observed across the five evaluated thresholds. Overall accuracy at this threshold was 73.5 percent.

Increasing the threshold produced the opposite effect. At a threshold of 0.70, precision increased to 91.67 percent and false positives decreased to only 1. However,

recall declined to 18.33 percent, and the model failed to identify 49 of the 60 actual bad-risk cases.

Figure 2 illustrates how accuracy, precision, recall, and F1 Score change across the evaluated classification thresholds.

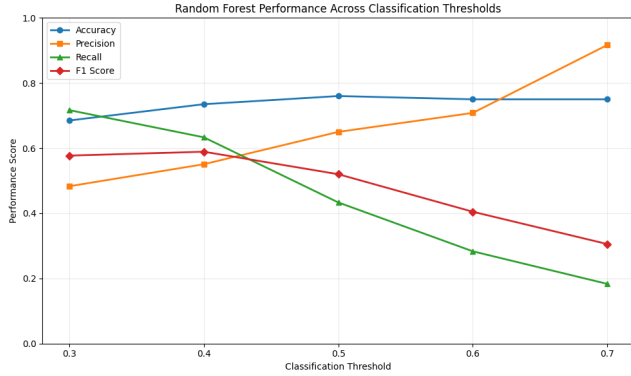


Figure 2: Random Forest performance across classification thresholds.

These results demonstrate that the classification threshold affects not only statistical performance but also the types of business errors produced by the model. A lower threshold makes the system more sensitive to potential bad-risk cases but increases the number of good-risk applicants incorrectly classified as bad risk. A higher threshold reduces false positives but increases the number of bad-risk applicants incorrectly classified as good risk.

The classification threshold should therefore be considered a business decision parameter rather than only a technical model setting. The appropriate threshold depends on the relative consequences of false positives and false negatives, organizational risk tolerance, and the objectives associated with the decision process.

5.4 Selective Automation and Human-Review Configuration

To examine how AI-based decision making could operate with different levels of human involvement, a selective automation configuration was developed using the predicted bad-risk probabilities generated by the Random Forest model. Rather than allowing the model to make every decision automatically, prediction probabilities were used to separate cases into AI-handled decisions and cases designated for human review.

Cases with a predicted bad-risk probability below 0.30 were automatically classified as good risk. Predictions between 0.30 and 0.60 were designated for human review because they represented less decisive predictions. Predictions above 0.60 were automatically classified as bad risk. The boundaries of 0.30 and 0.60 were used as illustrative experimental cutoffs to examine how an intermediate range of prediction confidence could be used for decision routing. They were not optimized as operating thresholds for real credit decisions.

The experimental decision rule was therefore:

$$p < 0.30 \rightarrow \text{AI: Low Risk}$$

$$0.30 \leq p \leq 0.60 \rightarrow \text{Human Review}$$

$$p > 0.60 \rightarrow \text{AI: High Risk}$$

Applying this configuration to the 200-observation test set resulted in 135 cases being handled automatically by AI and 65 cases being designated for human review. The AI automation rate was therefore 67.5 percent, while the human-review rate was 32.5 percent.

Among all 200 test cases, 111 were automatically classified as low risk, 24 were automatically classified as high risk, and 65 were assigned to the human-review category.

The accuracy of the 135 cases handled automatically by AI was 82.22 percent. Within these AI-handled cases, the system produced 94 true negatives, 7 false positives, 17 false negatives, and 17 true positives. This corresponds to 111 correct automated decisions and 24 incorrect automated decisions.

The human-review group contained 39 actual good-risk cases and 26 actual bad-risk cases. Consequently, 26 of the 60 actual bad-risk cases in the complete test set, or 43.33 percent, fell within the probability range designated for human review.

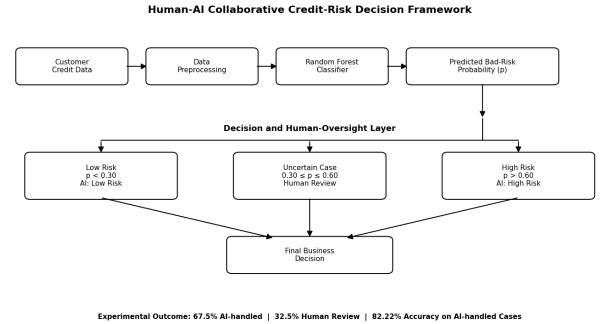


Figure 3: Selective automation and human-review framework.

These results show how predicted probabilities can be used to allocate different levels of decision authority within an AI-supported business process. Instead of requiring either complete human control or complete automation, the configuration allows more decisive model predictions to be processed automatically while reserving a selected range of less decisive cases for additional human assessment.

Importantly, the experiment did not involve actual human participants making decisions on the 65 review cases. The human-review category represents a simulated decision-allocation mechanism rather than measured human decision performance. The experiment therefore evaluates the performance of the AI-handled portion and the allocation of cases between automation and review, rather than the accuracy of a complete human-AI decision system.

5.5 Business Interpretation of the Experimental Results

The experimental results show that predictive performance alone does not provide a sufficient basis for determining the level of automated decision authority. Although Random Forest achieved the highest overall accuracy, its confusion matrix showed that substantial classification errors remained, particularly for actual bad-risk cases. The threshold analysis further demonstrated that these errors could be redistributed rather than eliminated by changing the classification boundary.

This trade-off has a direct business interpretation. Lower thresholds increased the identification of bad-risk cases but also increased the number of good-risk cases classified incorrectly. Higher thresholds reduced false positives but allowed more bad-risk cases to be classified as good risk. Selecting a classification threshold therefore requires consideration of the relative consequences of these errors rather than relying only on overall accuracy.

The selective automation experiment provides a further distinction between prediction and decision authority. Using the selected probability boundaries, 67.5 percent of the test cases were processed automatically, while 32.5 percent were designated for human review. Accuracy among the automatically handled cases was 82.22 percent, but 24 automated decisions remained incorrect. Restricting automation to more decisive predictions therefore reduced the number of cases exposed to automated decision making, but it did not eliminate classification errors.

The observed automation rate should not be interpreted as an optimal level of automation for credit-risk decisions. The probability boundaries were used to examine how prediction confidence can influence decision routing. Different error costs, organizational objectives, risk tolerances, and governance requirements could justify different boundaries and consequently different levels of automated processing.

Overall, the experiment indicates that AI decision autonomy is better treated as a configurable allocation of decision authority than as a choice between complete automation and complete human control. Predictive performance, uncertainty, and the consequences of classification errors provide technical evidence for that allocation, while the appropriate level of automation ultimately depends on the business context in which the model is used.

6 Discussion

The experimental findings provide further insight into the relationship between artificial intelligence, human decision making, and the level of autonomy that may be appropriate in business operations. Previous research has shown that AI can support organizational decision making through prediction, classification, information processing, and automation (Agrawal et al., 2019; Hasan, 2022). The present experiment adds an important distinction: the ability of a machine learning model to perform a prediction task does not by itself establish

that the resulting business decision should be transferred entirely to the system. Predictive reliability, the consequences of errors, uncertainty, and the organizational context in which a prediction is used all affect how much decision authority can reasonably be assigned to AI.

6.1 Predictive Performance and Decision Autonomy

The comparison of the five machine learning models shows that model selection involves trade-offs between different dimensions of predictive performance. Random Forest achieved the highest overall accuracy of 76.0 percent, SVM achieved the highest ROC-AUC of 0.7819, and Decision Tree produced the highest recall for bad-risk cases at 50.0 percent. No model therefore dominated across all of the measures considered.

This distinction matters when predictive output is connected to a business action. Aggregate measures such as accuracy indicate how frequently a model is correct overall, but they do not reveal whether the remaining errors are acceptable within a particular decision context (Paley et al., 2022). Research on AI investment similarly emphasizes that technological capabilities need to be connected to organizational problems and desired business outcomes rather than evaluated independently of their intended use (Gudigantala et al., 2023).

The Random Forest results illustrate this issue directly. Despite producing the highest accuracy, the model classified 34 of the 60 actual bad-risk cases as good risk at the default threshold. The result does not imply that Random Forest is unsuitable for credit-risk prediction. Instead, it shows that a model can perform comparatively well according to an aggregate measure while still producing errors that may matter substantially to the organization using it.

Decision autonomy should therefore not follow automatically from model performance. The relevant question is not simply whether a model achieves the highest accuracy, but whether its pattern of correct and incorrect predictions is acceptable for the decision authority being assigned to it.

6.2 Decision Thresholds and Business Risk

The threshold analysis shows that the behavior of an AI-based decision process is influenced not only by the trained model but also by the rule used to convert predicted probabilities into classifications. Changing the Random Forest threshold altered the balance between identifying bad-risk cases and incorrectly classifying good-risk cases.

At the default threshold of 0.50, recall for bad-risk cases was 43.33 percent. Reducing the threshold to 0.40 increased recall to 63.33 percent and reduced false negatives from 34 to 22, while lowering overall accuracy from 76.0 percent to 73.5 percent. At 0.30, false negatives declined further to 17, but false positives increased to 46.

These changes are not merely statistical differences. They represent different decision policies. A lower threshold places greater emphasis on identifying potential bad-risk cases, accepting that more good-risk cases will also be flagged. A higher threshold produces the opposite trade-off. Neither direction is inherently preferable without knowing the consequences attached to each type of error.

This finding reinforces the distinction between prediction and organizational decision making discussed by Agrawal et al. (2019). The model produces an estimate of risk, but the organization determines how that estimate should be translated into action. The classification threshold therefore becomes part of the business decision rule rather than simply a fixed technical setting.

This interpretation is also consistent with Shrestha et al. (2019), who show that AI-based organizational decision making involves choices about how decision authority is structured. In the present experiment, threshold selection represents one mechanism through which organizational priorities can influence how model predictions are converted into decisions.

6.3 Selective Automation and Human Review

The selective automation configuration provides an alternative to treating decision making as a choice between complete human control and complete AI autonomy. Instead of requiring the Random Forest model to make an automated decision for every case, predicted probabilities were used to determine whether a case would be processed automatically or designated for additional human review.

Using the experimental probability boundaries, 67.5 percent of the test cases were assigned to automated processing and 32.5 percent were designated for human review. Accuracy among the automatically handled cases was 82.22 percent. This result shows how organizations could reduce the number of cases requiring direct human attention without requiring the AI system to act autonomously on every prediction.

The configuration reflects the broader view of human-AI decision making described by Puranam (2021), in which the allocation of tasks depends on the relative capabilities of humans and algorithms rather than assuming that one must replace the other. It is also consistent with organizational decision structures described by Shrestha et al. (2019), where decision authority can be distributed between human and AI actors in different ways.

The main value of the experiment is therefore not the specific automation rate of 67.5 percent. That rate results from the probability boundaries selected for the experiment and should not be treated as an optimal level of automation. Instead, the experiment shows how prediction confidence can be incorporated into a decision-routing mechanism. More decisive predictions can be processed differently from observations that fall within a predefined uncertainty region.

Such an arrangement also relates to intelligent au-

tomation research, which emphasizes that automation needs to be considered together with human judgment, organizational processes, and governance (Coombs et al., 2020). Selective routing provides one practical mechanism through which those considerations can be incorporated into an AI-supported workflow.

6.4 Limits of Reduced Human Involvement

The selective automation results also show why confidence-based routing should not be interpreted as a guarantee of correct autonomous decision making. Among the 135 cases processed automatically, 24 were still classified incorrectly, including 17 false negatives and 7 false positives.

This is an important limitation because an incorrect prediction can occur even when the predicted probability falls outside the region designated for human review. Prediction confidence therefore provides information that can help allocate decision authority, but confidence alone does not establish that an automated decision is correct.

The human-review category also should not be interpreted as evidence that the resulting decisions would necessarily become correct. The experiment did not include human participants, and the 65 cases assigned to review were not subsequently evaluated by human decision makers. The experiment demonstrates a routing mechanism, not the measured performance of a complete human-AI decision system.

This distinction has implications for accountability. Delegating immediate decision execution to an AI system does not necessarily transfer responsibility for the consequences of that decision. Shrestha et al. (2019) identify this issue in organizational structures that allocate decision authority between humans and AI, while Tsamados et al. (2022) highlight broader concerns involving transparency, fairness, accountability, and the consequences of algorithmic decisions.

Human oversight therefore remains relevant for reasons beyond model uncertainty alone. Organizations must determine which errors are acceptable, how automated decisions can be monitored, when intervention should occur, and who remains responsible when an automated outcome causes harm or loss.

6.5 Business Value and the Appropriate Level of Autonomy

The findings also demonstrate why greater automation should not automatically be interpreted as greater business value. An AI system could technically process every case without direct human intervention, but doing so would expose the organization to all of the model's classification errors.

Selective automation changes this relationship by reducing the number of cases processed autonomously and reserving part of the workload for additional review. This may reduce the number of cases exposed to automated decision making, but it also retains a requirement

for human resources. The resulting trade-off is therefore not simply between more and less advanced technology. It is between different combinations of automation, labor, speed, error exposure, and oversight.

This interpretation is consistent with research showing that AI capabilities create value only when they are combined effectively with organizational resources and capabilities (Mikalef and Gupta, 2021). It also reflects the digitalization paradox discussed by Linde et al. (2021), in which investment in advanced technology does not automatically translate into successful business outcomes.

The appropriate level of autonomy must consequently be evaluated in relation to the objective of the business process. In one setting, reducing false negatives may justify greater human review or a lower classification threshold. In another, processing speed and operating cost may receive greater weight. Regulatory requirements, reputational consequences, customer impact, and organizational risk tolerance may further change the acceptable balance.

There is therefore no universal automation percentage that follows from the experimental results. The value of selective automation lies in allowing decision authority to be configured according to the requirements of the business context rather than maximizing automation as an objective in itself.

6.6 Implications for AI-Driven Business Operations

Taken together, the literature and experimental findings support understanding AI decision autonomy as a continuum. At one end, AI may provide predictions or recommendations while humans retain decision authority. At intermediate levels, some cases or decision components may be processed automatically while others are referred for human review. At higher levels of autonomy, AI may execute decisions directly when predictive reliability, error consequences, and governance conditions make such delegation acceptable.

The experiment provides a practical illustration of how movement along this continuum can occur. Model selection determines the underlying predictive behavior, threshold selection changes the balance between different classification errors, and confidence-based routing determines which cases are exposed to automated decision authority. These mechanisms show that autonomy is not simply a property of the machine learning model itself. It is also shaped by the decision rules and organizational controls surrounding the model.

This addresses the central gap identified in the literature review by connecting predictive performance with the allocation of operational decision authority. Research on human-AI collaboration, intelligent automation, organizational AI capability, and responsible AI provides the organizational foundation for this issue, while the experiment demonstrates how model behavior can affect that allocation in a measurable decision setting.

The central question for organizations is therefore not

simply whether AI is technically capable of generating a decision. A more consequential question is under what conditions an organization should allow an AI-generated prediction to produce an action without direct human intervention.

The findings suggest that this decision should consider predictive reliability, uncertainty, the consequences of different errors, expected business value, accountability, and the availability of appropriate oversight. Greater autonomy may be reasonable where these conditions can be managed effectively. Where uncertainty or potential consequences are greater, retaining human involvement may provide a more appropriate allocation of decision authority.

AI-driven transformation of business operations should therefore not be understood as the progressive removal of people from decision making. It is better understood as the redesign of decision authority around the respective capabilities, limitations, and responsibilities of humans and AI.

7 Limitations and Future Research

The findings of this study should be considered alongside several limitations. First, the experimental analysis is based on a single credit-risk dataset containing 1,000 observations. Credit risk provides a useful setting for examining AI-supported business decision making because different prediction errors can have different operational consequences. However, the findings cannot be assumed to generalize to all business decisions. Applications such as fraud detection, supply chain management, customer service, and resource allocation may involve different levels of uncertainty, risk, and requirements for human oversight.

Second, the experiment evaluates five supervised machine learning algorithms using a defined set of customer attributes. Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Gradient Boosting provide comparisons across different classification approaches, but they do not represent the full range of AI methods available to organizations. Different models may produce different patterns of predictive performance, uncertainty, and classification errors. Future research could evaluate additional approaches to determine whether the relationship between predictive reliability and the allocation of decision authority remains consistent across model types.

Third, the selective automation and human-review configuration was evaluated using probability boundaries of 0.30 and 0.60. These values were used to investigate how predictions with different levels of model confidence could be allocated between automated processing and human review. They should not be interpreted as optimal boundaries for real credit decisions. In practice, appropriate boundaries would depend on the relative costs and consequences of false positives and false negatives, organizational risk tolerance, regulatory requirements, and the cost and availability of human review.

A further limitation is that the experiment designates cases for human review but does not measure how humans would actually decide those cases. The 82.22 percent accuracy reported for AI-handled cases therefore represents only the performance of the automated portion of the configuration, not the accuracy of a complete human-AI decision system. Future research could conduct controlled experiments comparing human-only, AI-only, and combined human-AI decision processes using the same cases. Such experiments could evaluate accuracy, decision time, cost, consistency, and whether human reviewers are able to identify and correct errors that would otherwise be made by the model.

The experiment also does not provide a complete assessment of fairness or explainability. The dataset contains attributes including age and sex, but the analysis does not examine whether predictive performance or error rates differ across demographic groups. In consequential applications such as credit decisions, the use of particular attributes may also be restricted or require additional justification depending on the applicable legal and regulatory environment. Future research should therefore examine subgroup performance, appropriate fairness measures, model explanations, and the consequences of excluding or restricting particular attributes before greater decision authority is assigned to AI systems.

The analysis also represents a static classification setting in which models are trained and evaluated using historical data. Business conditions can change as customer behavior, economic circumstances, organizational policies, and data distributions change. A model that performs adequately under one set of conditions may become less reliable as those conditions shift (Thrall et al., 2018). Future research could therefore investigate continuous performance monitoring, distribution shift, periodic retraining, dynamic decision thresholds, and mechanisms for increasing human oversight when model reliability deteriorates (Paleyes et al., 2022).

Another limitation is that the held-out test set was used both to compare model performance and to conduct the subsequent threshold and selective-routing analyses. As a result, the threshold and routing results should be interpreted as exploratory rather than as independently validated operating policies. Future research could use separate training, validation, and test sets, or cross-validation combined with an untouched final test set, to select models and decision boundaries before evaluating the final configuration.

Future studies could also extend the experimental approach to larger datasets and additional business domains. Such work would help determine whether similar relationships between predictive confidence, classification errors, decision consequences, automation, and human involvement occur across different organizational settings. Rather than attempting to identify a universal level of AI autonomy, future research could develop and evaluate decision frameworks in which the degree of automation is adjusted according to the characteristics, uncertainty, risk, and consequences of individual business decisions.

8 Managerial and Practical Implications

The findings have several implications for organizations considering greater use of AI in operational decision making. First, predictive accuracy should not be treated as the sole criterion for determining whether a business decision is suitable for automation. In the experiment, Random Forest achieved the highest overall accuracy at 76.0 percent but still classified 34 of the 60 actual bad-risk cases as good risk at the default threshold. Managers should therefore examine the types and consequences of model errors rather than relying only on aggregate performance measures.

The relative importance of false positives and false negatives should be evaluated in relation to the objectives of the business process. The threshold analysis showed that reducing one type of error can substantially increase the other. At a threshold of 0.30, false negatives decreased to 17 while false positives increased to 46. At a threshold of 0.70, false positives decreased to one while false negatives increased to 49. The appropriate operating threshold should therefore reflect the organization’s tolerance for different errors and the financial, operational, or other consequences associated with them. This is consistent with the view that AI investments should be connected to clearly defined organizational problems and desired business outcomes (Gudigantala et al., 2023).

Second, organizations do not necessarily need to choose between complete manual processing and complete automation. The selective automation and human-review configuration shows an intermediate approach in which decision authority can be allocated according to prediction confidence. In the experiment, 67.5 percent of the test cases were processed automatically, while 32.5 percent were designated for human review. Accuracy among the AI-handled cases was 82.22 percent.

In practice, this type of configuration could allow organizations to concentrate human attention on cases for which the model provides less decisive predictions rather than requiring the same level of human involvement in every case. This reflects research describing human and AI capabilities as potentially complementary rather than treating AI solely as a substitute for human work (Puranam, 2021; Rai et al., 2019).

Managers should not, however, interpret prediction confidence as a guarantee that an automated decision is correct. The AI-handled portion of the experiment still contained 17 false negatives and 7 false positives. Automated decisions therefore require monitoring even when they are restricted to more decisive predictions. Organizations adopting this approach should establish procedures for performance monitoring, periodic evaluation, escalation, and intervention when model behavior becomes unacceptable (van Giffen et al., 2022). These controls should also account for security threats to deployed machine-learning systems, including data poisoning, model stealing, model inversion, and other adversarial manipulations (Kumar et al., 2020).

Accountability also needs to remain explicit as decision authority is delegated to AI. An organization may allow an algorithm to produce and execute a decision without immediate human approval, but responsibility for the consequences does not automatically transfer to the technology (Shrestha et al., 2019). Organizations should therefore define responsibility for model development, validation, deployment, monitoring, human review, and intervention. These responsibilities become increasingly important as direct human involvement in individual decisions decreases.

A practical approach is to determine the level of AI autonomy according to the characteristics of each decision process. Decisions made under relatively predictable conditions, with reliable model performance and manageable consequences of error, may support greater automation. Decisions involving greater uncertainty, unusual circumstances, substantial consequences, or stronger accountability requirements may justify greater human involvement.

Organizations can therefore treat AI adoption as a gradual allocation of decision authority rather than a one-time transition to full automation. AI may initially provide predictions or recommendations while humans retain final authority. Selected decisions can subsequently be automated when sufficient evidence supports doing so, while uncertain or higher-risk cases continue to receive human review. Ongoing performance monitoring can then provide evidence for increasing, maintaining, or reducing the level of automation.

The managerial objective should therefore not be to maximize the percentage of decisions handled by AI. It should be to determine where automation creates sufficient operational value while keeping the consequences of prediction errors manageable and maintaining organizational accountability. Human involvement remains valuable where judgment, contextual understanding, intervention, or responsibility is required, while AI can be assigned greater decision authority where its performance and the consequences of error make such delegation appropriate.

9 Conclusion

Artificial intelligence is expanding the ability of organizations to automate activities that previously required direct human participation. Machine learning systems can analyze large quantities of information, identify patterns, generate predictions, and support operational decisions at a speed and scale that would be difficult to achieve through manual processes alone. These capabilities create opportunities for greater automation, but they also raise an important question concerning how much decision authority should be transferred from people to AI systems.

This study examined that question through a review of literature on AI in business, human-AI decision making, automation, business value, and responsible governance, together with an experimental analysis of machine-learning-based credit-risk classification. Five

supervised machine learning algorithms were evaluated using a dataset containing 1,000 observations. The models produced different trade-offs between accuracy, precision, recall, F1 Score, and ROC-AUC, showing that the suitability of a model for business decision making cannot be determined from a single performance measure.

Random Forest achieved the highest overall accuracy at 76.0 percent, while SVM achieved the highest ROC-AUC at 0.7819. However, the error analysis showed that comparatively strong aggregate performance did not eliminate important classification errors. At the default threshold, Random Forest incorrectly classified 34 of the 60 actual bad-risk cases as good risk.

The threshold analysis shows that the behavior of an AI-based decision process is influenced not only by the trained model but also by the rule used to convert predicted probabilities into classifications. Lower thresholds increased the identification of bad-risk cases but also increased the number of good-risk cases classified incorrectly. Threshold selection should therefore reflect the relative consequences of different errors rather than being treated only as a technical setting.

The selective automation and human-review configuration provided an alternative to requiring either complete human control or complete automation. Using predicted probabilities to route cases, 67.5 percent of the test observations were processed automatically, while 32.5 percent were designated for human review. Accuracy among the AI-handled cases was 82.22 percent. The experiment therefore shows how prediction confidence can be used to allocate different levels of decision authority within a business process.

Selective automation, however, did not eliminate classification errors. Among the automatically processed cases, 17 actual bad-risk observations were still incorrectly classified as good risk. The human-review cases were also not evaluated by actual human participants. The findings therefore do not establish the performance of a complete human-AI decision system or identify an optimal level of automation. Instead, they show that reducing direct human involvement requires consideration of both model behavior and the consequences of the decisions being automated.

The study consequently views AI decision autonomy as a continuum. Some decisions may remain under human control, others may combine automated processing with human review, and some may permit greater autonomous operation. The appropriate position on this continuum depends on predictive reliability, uncertainty, the consequences of errors, organizational objectives, business value, accountability requirements, and the availability of effective oversight.

The central question for organizations is therefore not simply whether AI can make a business decision, but when an AI-generated prediction should be permitted to produce a business action without direct human intervention. Addressing this question requires technical evaluation to be considered together with business judgment and responsible governance. AI-driven transformation should be understood not as the removal of

humans from business decision making, but as the deliberate allocation of decision authority between humans and AI according to the requirements and consequences of each decision.

References

- Nasiba Mahdi Abdulkareem and Adnan Mohsin Abdulazeez. Machine learning classification based on random forest algorithm: A review. *International Journal of Science and Business*, 5(2):128–142, 2021. doi: 10.5281/zenodo.4471118.
- Ajay Agrawal, Joshua Gans, and Avi Goldfarb. Prediction, judgment, and complexity: A theory of decision-making and artificial intelligence. In Ajay Agrawal, Joshua Gans, and Avi Goldfarb, editors, *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press, 2019.
- Nicholas Berente, Bin Gu, Jan Recker, and Radhika Santhanam. Managing artificial intelligence. *MIS Quarterly*, 45(3):1433–1450, 2021. doi: 10.25300/MISQ/2021/16274.
- Erik Brynjolfsson, Daniel Rock, and Chad Syverson. Artificial intelligence and the modern productivity paradox: A clash of expectations and statistics. Technical Report 24001, National Bureau of Economic Research, 2017.
- Crispin Coombs, Donald Hislop, Stanimira K. Taneva, and Sarah Barnard. The strategic impacts of intelligent automation for knowledge and service work: An interdisciplinary review. *The Journal of Strategic Information Systems*, 29(4):101600, 2020. doi: 10.1016/j.jsis.2020.101600.
- Dimitris C. Gkikas and Prokopis K. Theodoridis. Ai in consumer behavior. In Maria Virvou et al., editors, *Advances in Artificial Intelligence-based Technologies*, volume 22 of *Learning and Analytics in Intelligent Systems*, pages 147–176. Springer Nature Switzerland, 2022.
- Naveen Gudigantala, Sreedhar Madhavaram, and Pelin Bicen. An ai decision-making framework for business value maximization. *AI Magazine*, 44(1):67–84, 2023. doi: 10.1002/aaai.12076.
- Ahmed Rizvan Hasan. Artificial intelligence (ai) in accounting & auditing: A literature review. *Open Journal of Business and Management*, 10(1):440–465, 2022. doi: 10.4236/ojbm.2022.101026.
- Mohammad Hossein Jarrahi. Artificial intelligence and the future of work: Human–ai symbiosis in organizational decision making. *Business Horizons*, 61(4):577–586, 2018. doi: 10.1016/j.bushor.2018.03.007.
- Muhammad Adnan Khan, Shazia Saqib, Tahir Alyas, Anees Ur Rehman, Yousaf Saeed, Asim Zeb, Mahdi Zareei, and Ehab Mahmoud Mohamed. Effective demand forecasting model using business intelligence empowered with machine learning. *IEEE Access*, 8:116013–116023, 2020. doi: 10.1109/ACCESS.2020.3003790.
- Ram Shankar Siva Kumar, Magnus Nyström, John Lambert, Andrew Marshall, Mario Goertzel, Andi Comis-soneru, Matt Swann, and Sharon Xia. Adversarial machine learning–industry perspectives. In *2020 IEEE Security and Privacy Workshops (SPW)*, pages 69–75. IEEE, 2020. doi: 10.1109/SPW50608.2020.00028.
- Lina Linde, David Sjödin, Vinit Parida, and Heiko Gebauer. Evaluation of digital business model opportunities: A framework for avoiding digitalization traps. *Research-Technology Management*, 64(1):43–53, 2021. doi: 10.1080/08956308.2021.1842664.
- Patrick Mikalef and Manjul Gupta. Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. *Information & Management*, 58(3):103434, 2021. doi: 10.1016/j.im.2021.103434.
- Shrutika Mishra and A. R. Tripathi. Ai business model: An integrative business approach. *Journal of Innovation and Entrepreneurship*, 10(1):18, 2021. doi: 10.1186/s13731-021-00157-5.
- Andrei Paleyes, Raoul-Gabriel Urma, and Neil D. Lawrence. Challenges in deploying machine learning: A survey of case studies. *ACM Computing Surveys*, 55(6):1–29, 2022. doi: 10.1145/3533378.
- Nikhil Patel and Sandeep Trivedi. Leveraging predictive modeling, machine learning personalization, nlp customer support, and ai chatbots to increase customer loyalty. *EQM ResearchBerg*, 3(3):1–24, 2020.
- Phanish Puranam. Human–ai collaborative decision-making as an organization design problem. *Journal of Organization Design*, 10(2):75–80, 2021. doi: 10.1007/s41469-021-00095-2.
- Arun Rai, Panos Constantinides, and Saonee Sarker. Next-generation digital platforms: Toward human–ai hybrids. *MIS Quarterly*, 43(1):iii–ix, 2019.
- Wiebke Reim, Josef Åström, and Oliver Eriksson. Implementation of artificial intelligence (ai): A roadmap for business model innovation. *AI*, 1(2):180–191, 2020. doi: 10.3390/ai1020011.
- Krunoslav Ris, Zeljko Stankovic, and Zoran Avramovic. Implications of implementation of artificial intelligence in the banking business with correlation to the human factor. *Journal of Computer and Communications*, 8:130–144, 2020. doi: 10.4236/jcc.2020.811010.
- Rainer Schmidt, Alfred Zimmermann, Michael Moehring, and Barbara Keller. Value creation in connectionist artificial intelligence – a research agenda. In *Proceedings of the Americas Conference on Information Systems (AMCIS 2020)*, 2020.

- Yash Raj Shrestha, Shiko M. Ben-Menahem, and Georg von Krogh. Organizational decision-making structures in the age of artificial intelligence. *California Management Review*, 61(4):66–83, 2019. doi: 10.1177/0008125619862257.
- Bernd Carsten Stahl, Andreas Andreou, Philip Brey, Tally Hatzakis, Alexey Kirichenko, Kevin Macnish, Stephanie Laulhé Shaelou, Andrew Patel, Mark Ryan, and David Wright. Artificial intelligence for human flourishing—beyond principles for machine learning. *Journal of Business Research*, 124:374–388, 2021. doi: 10.1016/j.jbusres.2020.11.030.
- James H. Thrall, Xiang Li, Quanzheng Li, Cinthia Cruz, Synho Do, Keith Dreyer, and James Brink. Artificial intelligence and machine learning in radiology: Opportunities, challenges, pitfalls, and criteria for success. *Journal of the American College of Radiology*, 15(3): 504–508, 2018. doi: 10.1016/j.jacr.2017.12.026.
- Andreas Tsamados, Nikita Aggarwal, Josh Cows, Jessica Morley, Huw Roberts, Mariarosaria Taddeo, and Luciano Floridi. The ethics of algorithms: Key problems and solutions. *AI & Society*, 37:215–230, 2022. doi: 10.1007/s00146-021-01154-8.
- Benjamin van Giffen, Dennis Herhausen, and Tobias Fahse. Overcoming the pitfalls and perils of algorithms: A classification of machine learning biases and mitigation methods. *Journal of Business Research*, 144:93–106, 2022. doi: 10.1016/j.jbusres.2022.01.076.
- Gioia Volkmar, Peter M. Fischer, and Sven Reinecke. Artificial intelligence and machine learning: Exploring drivers, barriers, and future developments in marketing management. *Journal of Business Research*, 149: 599–614, 2022. doi: 10.1016/j.jbusres.2022.04.007.
- Serge-Lopez Wamba-Taguimdje, Samuel Fosso Wamba, Jean Robert Kala Kamdjoug, and Chris Emmanuel Tchatchouang Wanko. Influence of artificial intelligence (ai) on firm performance: The business value of ai-based transformation projects. *Business Process Management Journal*, 26(7):1893–1924, 2020. doi: 10.1108/BPMJ-10-2019-0411.
- John Zhang, Ming Fan, Bin Gu, Vijay Mookerjee, Bin Zhang, and J. Leon Zhao. Business values/implications of ai and machine learning. *Data and Information Management*, 2(3):121–129, 2018. doi: 10.2478/dim-2018-0016.